

PRUTHVIRAJ SHINDE

Machine Learning Engineer | AI & Data Systems | Clean Energy & Sustainability AI

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PROFESSIONAL SUMMARY

Machine Learning engineer and Data Science student with hands-on experience designing end-to-end AI pipelines, predictive systems, and deep learning models across real-world domains. Proven track record of owning problems from data preprocessing through model deployment — including transfer learning studies achieving >95% accuracy and production-grade NLP systems deployed at scale. Passionate about applying ML to sustainability-driven challenges such as forecasting, intelligent data pipelines, and scalable AI systems. Brings a bias toward action, rigorous experiment documentation, and a growth mindset suited to fast-moving, mission-driven startup environments.

KEY SKILLS

Languages & Libraries: Python, SQL, R | NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

ML & Deep Learning: Regression, Classification, Clustering, XGBoost, Random Forest, TensorFlow, PyTorch, ResNet-50, DenseNet-121, Transfer Learning, Fine-Tuning

AI & NLP: NLP Pipelines, LangChain, LangGraph, RAG (Retrieval-Augmented Generation), LLM Integration, Azure OpenAI, Gemini AI

Data Engineering: Data Cleaning, EDA, Feature Engineering, Jupyter Notebooks, Reproducible Workflows, Streamlit

Cloud & Tools: Google Cloud Platform, AWS, Azure, Git, GitHub, VS Code, Power Automate, Supabase

Core Concepts: Deep Learning, Probability & Statistics, Algorithms, DBMS, Data Structures, Time Series (Exposure), Reinforcement Learning (Exposure)

PROFESSIONAL EXPERIENCE

Tata Power [\[Internship Certificate\]](#) May 2025 – Jul 2025 *AI / Machine Learning Intern* Mumbai, Maharashtra

- Built an end-to-end AI data pipeline for automated PDF extraction and vendor evaluation using Azure OpenAI and RAG, integrating Excel auto-fill dashboards that reduced manual data processing time by ~70% and improved procurement efficiency.
- Developed a context-aware ML inference system using Retrieval-Augmented Generation (RAG) to process unstructured RFQ documents, improving quotation accuracy and decision consistency across enterprise procurement workflows.
- Contributed to reproducible ML workflows by documenting pipeline architecture, experiment configurations, and findings for cross-functional engineering teams — enabling iterative improvement and knowledge transfer.
- Collaborated with senior engineers and business stakeholders to validate outputs and present ML-driven insights in team discussions, bridging technical results with operational decisions.

Gocrakit [\[Internship Certificate\]](#) Apr 2025 – Jul 2025 *Machine Learning Intern* Bangalore, Karnataka

- Engineered an AI + RAG-based resume parsing and intelligent recommendation system using LLM-powered NLP pipelines, automating candidate screening and improving recommendation relevance by ~35%.
- Assisted in designing scalable ML system architectures, building and testing NLP-driven data pipelines, and exploring modelling approaches for personalization and automated decision-support — gaining exposure to production-grade ML systems.
- Maintained clear documentation of modelling approaches, preprocessing steps, and evaluation results, contributing to reproducible experiment workflows for team review.
- Shared data insights and model performance results in team discussions, developing strong communication habits around ML experimentation and iterative model refinement.

PROJECTS

Transfer Learning Study: ResNet-50 vs. DenseNet-121 on CalTech-101 | PyTorch, TensorFlow, Python [\[GitHub\]](#) Oct 2025 – Dec 2025

- Designed and executed a comprehensive transfer learning study comparing ResNet-50 (~23.7M params) and DenseNet-121 (~7.1M params) using ImageNet-pretrained weights, achieving >95% Top-1 accuracy across all four model-strategy combinations on CalTech-101.
- Engineered a novel Adaptive Fine-Tuning (AFT) strategy using importance-based gradient masking on the top 25% most discriminative filters, enabling ResNet-50 to match DenseNet Standard Fine-Tuning performance with 6× fewer trainable parameters (~3.8M vs ~22.3M) — demonstrating strong model efficiency and reproducible experiment design.

ML-Based Diabetes Risk Prediction System | XGBoost, Scikit-learn, Streamlit, Python [\[GitHub\]](#) Oct 2025 – Dec 2025

- Designed and trained a diabetes risk prediction model using XGBoost and hybrid scoring, applying data preprocessing, feature engineering, and cross-validation to achieve ~90% classification accuracy on patient health data.
- Built a Streamlit real-time inference application with probability-based predictions and visual risk dashboards — implementing a fully reproducible ML workflow with documented evaluation metrics and modular pipeline design.

CyberGuard AI — Multilingual AI Complaint Intelligence Platform | Python, Gemini AI, NLP, Supabase, Streamlit [\[GitHub\]](#)
Jan 2025 – Apr 2025

- Led full-stack development of a production-grade AI classification system supporting 22 Indian languages via TTS/STT NLP pipelines, achieving 95% classification accuracy across 100+ test scenarios.
- Designed complete data architecture — Supabase NoSQL storage, Gemini AI classification, and SMTP notification layer — demonstrating end-to-end ownership from data preprocessing through deployment, reducing manual data entry by ~70%.

EDUCATION

Vidyashilp University

Aug 2023 – Jun 2027

B.Tech in Computer Science Engineering (Data Science) — CGPA: 8.41 / 10.0 | Bangalore, Karnataka

CERTIFICATIONS & ACHIEVEMENTS

Certifications: [Accenture: Data Analytics & Visualization](#) | [Deloitte: Data Analytics](#) | [BCG X: GenAI](#)

Recognition: ISRO Bharatiya Antariksh | [AI Rising](#) | [Hackstreet](#)